

DIVYESH MISTRY

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EDUCATION

University of Southern California

Master of Science in Computer Science – Data Science; GPA: **3.67/4.00**

Los Angeles, CA

Aug 2024 – May 2026

University of Mumbai

Bachelor of Engineering in Computer Engineering (Data Science); GPA: **3.7/4.00**

Mumbai, India

Aug 2021 – May 2024

EXPERIENCE

Data Scientist Intern

June 2025 – Aug 2025

ServiceLink

Irving, TX

- Optimized post-close document analysis by 38%+ by fine-tuning **Gemma-3-4b-it** via **LoRA** and **PEFT** on a custom **4,000+ image dataset**.
- Achieved optimal model architecture selection for **GPU-accelerated inference** by benchmarking **10+ vision-language** models, evaluating accuracy, **memory bandwidth**, inference speed, and model size.
- Reduced model footprint by **70%** via **4-bit quantization** (INT4), enhancing kernels for resource-constrained **GPU environments** leveraging **LLaMA.cpp**.
- Delivered **real-time**, on-device document analysis across 3+ platforms, deploying a **React Native** app combined with **llama.rn** for 100% data privacy.

Software Engineer

Dec 2024 – May 2025

USC Ming Hsieh Department of Electrical and Computer Engineering

Los Angeles, CA

- Automated an **end-to-end** epidemiological data ingestion and **prediction pipeline**, reducing weekly forecast generation time to **under 20 seconds** by integrating real-time healthcare datasets with a custom **Random Forest architecture**.
- Engineered forecasting architecture to support **parallel processing**, enabling efficient, simultaneous execution of machine learning projections across **50+ regions**.
- Analyzed and resolved training and **inference performance bottlenecks** in a Python-based simulation, re-engineering core modules with **Cython** to achieve a **25% speedup** on high-performance compute nodes.

Data Analyst Intern

Aug 2023 – Oct 2023

OziBook

Mumbai, India

- Led data workflows for a **30-member cross-functional team**, building a centralized client database that successfully converted **2** new enterprise clients.
- Evaluated **900+ potential client profiles** through **Pandas** and **Tableau** to identify and shortlist high-conversion targets aligned with company services.
- Visualized complex datasets into actionable business stories via **Tableau** and **Power BI** dashboards.

PROJECTS

AI-Powered Academic Research Assistant | [Link](#)

- Built a scalable **NLP framework** to transform **unstructured academic text** into **structured summaries**, utilizing **Parallel Processing** to cut down data ingestion and summarization latency by **90%**.
- Fine-tuned **LongT5 Transformer model** for summarization using a dataset of **1,000** computer science research papers.
- Consolidated a hybrid summarization approach by applying **LexRank** algorithm to identify Key Takeaways, decreasing manual literature review time by **90%** through automated **extractive-abstractive** filtering.

AutoML Command Center | [Link](#)

- Built a full-stack AutoML platform using **Next.js**, **FastAPI**, **PostgreSQL**, **Redis**, and **Celery**, supporting **5+** end-to-end workflows including **dataset upload**, **profiling**, **experiment tracking**, **model training**, **deployment**, and **real-time inference**, cutting down manual ML pipeline setup time by **70%**.
- Developed an **AI-powered, dataset-aware assistant** with persistent chat history and tool-driven orchestration, motorizing **80%+** of experiment guidance and workflow execution across training, deployment, and prediction tasks.
- Deployed platform to production leveraging **Vercel** and **Fly.io** with managed **Postgres** and **Redis**, improving system reliability by **30%+** through cross-validation, lightweight hyperparameter tuning, telemetry tracking, and prediction history dashboards, while enabling sub-second inference latency.

TECHNICAL SKILLS

Languages: Python (Cython), R, Java, C++, C#, SQL, JavaScript (TypeScript), Scala, Dart, Bash.

Machine Learning & AI: PyTorch, TensorFlow, Keras, Scikit-learn, HuggingFace, LangChain, LLM Fine-Tuning (LoRA/PEFT), NLP, Computer Vision (OpenCV), Quantization, XGBoost, SpaCy, Tesseract.

Data Engineering & Cloud: AWS, GCP, Azure, Apache Spark, ETL Pipelines, PostgreSQL, NoSQL (MongoDB, Cassandra, Firebase), Vector Databases, Redis, Kafka, Data Warehousing, Parallel Processing, Data Mapping/Migration.

Frameworks & Tools: Flask, Django, StreamLit, Next.js, React Native, Flutter, Tableau, Power BI, Git, Excel.